TRAVEL*FLOW*

J O U R N E Y B E Y O N D T H E H O R I Z O N

STAKEHOLDERS IDENTIFICATION TABLE

1. Internal Stakeholders

Stakeholder	Role / Responsibility	Importance	Influence	Interests / Positive Impacts	Concerns
Owner / CEO	Leads TravelFlow's overall strategy, business model, and financial direction. Final authority on product scope, pricing, and route expansion decisions.	**High**	**High**	Platform profitability, brand growth, expansion along the Albanian Riviera (Tiranë–Sarandë corridor), competitive positioning.	High infrastructure costs, driver compliance, legal licensing requirements, long-term system reliability.
Platform Administrator	Operates the admin dashboard daily: registers buses and rental cars, monitors all bookings and revenue, manages maintenance schedules, and oversees the 6-tab Finance Panel.	**High**	**High**	Centralised back-office control, automated KPI tracking (bookings, revenue, rentals), full GPS fleet visibility.	Data accuracy in localStorage, cybersecurity exposure, system downtime risk, scalability constraints.
Bus Driver	Accesses a dedicated Driver Portal on login. Validates passenger QR tickets via device camera (html5-qrcode), views assigned route/bus/passenger count, logs boarding sessions with timestamps.	**High**	**Medium**	Paperless boarding process, clear per-passenger seat and stop info, real-time validation results (Valid / Already Scanned / Not Found).	Device and camera reliability on remote mountain roads (Llogarë Pass), connectivity gaps, low-end device performance.
Project Manager / Team Lead	Coordinates the development sprint, sets milestones, resolves technical and organisational blockers, and ensures on-time delivery of all TravelFlow components.	**High**	**High**	Clear feature scope, team alignment across GPS engine, booking flow, and admin panel, on-schedule delivery.	Scope creep, shifting requirements, integration complexity across Google Maps, EmailJS, and QR libraries.
Development Team **(Etna Gjika, Tristan Halo, Dhimitri Shallo)**	Core SME reviewers and builders of TravelFlow. Collectively review and approve all 28 functional requirements (FR-01–FR-28). Responsible for GPS engine, booking flows, driver portal, autoTrips, maintenance scheduler, and finance panel.	**High**	**High**	Clean maintainable vanilla JS codebase, accurate GPS road-path engine (48 hairpins on Llogarë), reliable QR validation, autoTrips generation under 500ms.	Third-party API instability, browser compatibility across vanilla stack, localStorage limitations, concurrent booking edge-cases.
Academic Supervisor / Professor	Oversees the Software Engineering Analysis & Design course project. Evaluates all deliverables: requirements specification, user scenarios, user stories, stakeholder analysis, and final platform demonstration.	**High**	**High**	Rigorous academic methodology, complete SEAD documentation, traceability from US to FR to scenarios, professional presentation.	Incomplete non-functional coverage, missing stakeholder identification, poor requirement traceability, undocumented design decisions.
Fleet Maintenance Technician	Responsible for physical servicing of the 6-bus fleet and 12 rental cars. Uses the TravelFlow Maintenance module (admin panel) to log completed services; system auto-recalculates next due dates.	**Medium**	**Low**	Digital service records, automated next-service scheduling, overdue alerts per vehicle, bulk 'Mark All as Serviced' action.	Tracking accuracy of service intervals, reminder reliability, integration with external garage parts suppliers.

2. External Stakeholders

Stakeholder	Role / Responsibility	Importance	Influence	Interests / Positive Impacts	Concerns
Registered Passenger (Client)	Core platform user: registers with 6-digit email OTP verification, searches trips by route/date, books seats via visual seat map, pays online with Luhn-validated card, receives QR e-ticket by email, manages bookings and rentals via My Account.	**High**	**High**	Seamless booking (visual seat map + QR e-ticket + email delivery), fare transparency per stop segment, digital rental history, instant booking confirmation toast.	Personal data stored in localStorage (no server encryption), payment security, cancellation policy clarity, seat availability during peak season.
Car Rental Customer	Books one of 12 rental vehicles (Economy to Off-Road) via the platform. Selects pickup and return dates with live price calculation. Rented car is GPS-tracked on admin map and auto-drives back to base on rental expiry.	**High**	**Medium**	Freedom to explore Riviera at own pace, transparent daily pricing, GPS-secured vehicle, confirmed/returning/completed status tracking in My Account.	Insurance and damage dispute clarity, return process, limited vehicle variety vs. larger rental operators.
Guest / Unregistered User	Browses routes, destination highlights, and the Tourist Guide section without an account. Primary conversion target to registered client.	**Medium**	**Low**	Access to Riviera destination info (Sarandë, Himarë, Llogarë, Butrint, Ksamil) and route prices without mandatory registration.	Friction from mandatory sign-up to book; no ability to save searches, compare trips, or receive schedule alerts.
Albanian Riviera Tourist	Domestic or international traveller targeting coastal destinations (Sarandë, Dhërmi, Borsh, Himarë, Ksamil). Primary revenue driver during June–September peak season.	**High**	**Medium**	Rich tourist guide with destination highlights, easy multi-stop journey planning, real-time GPS bus tracking, direct booking without travel agents.	Language barriers if interface is Albanian-only, seasonal route limitations, connectivity in remote coastal areas.
DPSHTRR (Road Transport Directorate)	Albanian authority for vehicle registration, roadworthiness certificates, and intercity route operating permits. TravelFlow's 6-bus fleet must comply with all DPSHTRR mandates and inspection schedules.	**High**	**High**	Digital maintenance records aligned with inspection schedules, reduced administrative burden for permit renewals.	Regulatory non-compliance risk if maintenance logs are inaccurate, license renewal timelines, documentation overhead.
Ministria e Infrastrukturës dhe Energjisë	Sets national transport policy, approves intercity routes, and regulates passenger fares on Albanian highways (SH2, A1, SH8 Rivierë). TravelFlow's 6 routes all require ministerial route authorisation.	**High**	**High**	Alignment with national transport digitalisation strategy, route legality on Tiranë–Sarandë corridor, contribution to tourism infrastructure.	Policy changes affecting permitted stops or pricing caps, bureaucratic approval delays for new routes.
Komisioneri për Mbrojtjen e të Dhënave Personale	Albanian Data Protection Commissioner enforcing Law No. 124/2024 on Personal Data Protection (GDPR-aligned). Governs all personal data collected and stored by TravelFlow, including names, emails, booking codes, and payment details.	**High**	**High**	Privacy-by-design compliance, user trust, legal operation of the platform.	localStorage stores personal data client-side without encryption — direct compliance risk. No server-side GDPR controls currently implemented.

Stakeholder	Role / Responsibility	Importance	Influence	Interests / Positive Impacts	Concerns
AKEP (Electronic Communications Authority)	Regulates digital service providers and telecommunications infrastructure in Albania. Relevant for EmailJS email delivery services and GPS data transmission via Google APIs.	**Medium**	**Medium**	Compliant digital communications infrastructure for transactional email (OTPs, e-tickets) and GPS data flows.	Potential restrictions on third-party API data transmission, digital service registration requirements.
Google Maps / Directions API	Critical technology partner. Powers the entire GPS engine: fetches real road paths for all 6 routes via Directions API on load, traces every curve on Qafa e Llogarasë (48 hairpin turns, 1043m altitude), updates bus/car positions every 2 seconds.	**High**	**High**	Accurate real-road vehicle tracking (not straight-line), live bus markers with MOVING/STOPPED status, admin GPS map for all 18 vehicles simultaneously.	API quota exhaustion, cost increases with scale, third-party uptime dependency, GPS accuracy degradation in coastal valleys.
EmailJS (Email Service Provider)	Handles all transactional emails: 6-digit OTP verification codes on registration, and full e-ticket delivery (with barcode/QR) after booking confirmation. No server-side email infrastructure required.	**High**	**High**	Automated professional email delivery, verified-only accounts (no throwaway signups), instant ticket receipt to passenger inbox.	Spam filter interference, daily send-limit caps, template delivery failures, SDK version deprecation risk.
Payment & Card Validation Layer	Client-side Luhn algorithm validates card numbers (Visa, Mastercard, Amex) with live brand detection and logo display as the user types. CVV and expiry validation run before booking confirmation.	**High**	**Medium**	Real-time card validation feedback, no rejected cards at checkout, brand logo display builds user confidence.	Purely client-side validation is not PCI-DSS compliant — critical gap for production deployment at scale.
Third-Party Garages & Parts Suppliers	External mechanics and spare-parts providers servicing TravelFlow's 6-bus fleet and 12 rental cars. Their completed work is logged in the TravelFlow Maintenance module by the technician or admin.	**Medium**	**Low**	Timely service completion reflected digitally, auto-recalculated next-service dates, reduced unplanned fleet downtime.	Parts price fluctuations, geographic availability near Riviera routes, service quality inconsistency affecting fleet safety.
Competing Transport Operators	Other intercity bus companies operating on overlapping Albanian Riviera routes (Tiranë–Sarandë, Tiranë–Himarë). Drive competitive pressure on pricing, scheduling, and digital UX.	**Medium**	**Low**	Benchmarking drives TravelFlow to improve features (GPS tracking, QR boarding, seat maps) and pricing competitiveness.	Price undercutting on shared routes, loyalty erosion if competitors adopt digital ticketing, feature parity pressure.